Chloride-conductance calibration: sampling plan and laboratory cost options

Steamboat Hills geothermal outflow, Steamboat Creek (SBRR and SBGG)

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2026-10-29

Summary

I am calibrating continuous specific conductance from two in-stream loggers (SBRR at Rhodes Road above the thermal inflow, and SBGG at Geiger Grade below it) against laboratory chloride. **Method:** paired grab samples bracket logger readings; chloride (and major ions, alkalinity, field pH/temperature/conductance) is measured at the lab, and I fit time-aware models (linear, GAM, random forest) validated with rolling-origin cross-validation. **Application:** the calibrated record gives continuous chloride flux, which is the standard tracer for thermal-water discharge at Steamboat (Sorey and Spielman 2008, 2017). It lets us ask whether the 2025 Lower Sinter Terrace eruption changed outflow, and it replaces the sparse, expensive chemistry that has limited the discharge record. **Reasoning:** a Monte Carlo test on the real logger record (with the days the loggers sat in open air removed) shows daily sampling gives a reconstruction error of about 3-4% of mean conductance, while biweekly sampling gives about 8-10%, and 7-13% depending on the start day, so a short intensive window followed by monthly samples across a full year captures both fast variability and seasonal stability. **Hypothesis:** one site-specific relationship between conductance and chloride holds through the year, with deviations during storms and snowmelt from dilution. **Value:** once calibrated, the loggers give a continuous chloride and thermal-discharge record for a small fraction of the cost of regular chemistry, and the same samples provide the end-member chemistry for the mixing and reactive-transport work.

Sampling plan

Phase	Sites	Cadence and timing	What is measured
Intensive calibration	SBRR, SBGG	Daily (or sub-daily) for 5, 7 or 14 days; start fall/winter 2026	Cl, anions, cations, alkalinity (package set below); field pH, temperature, conductance
Monthly seasonal sampling	SBRR, SBGG	Once a month for 5 or 12 months after the intensive window	Same package
Storm / snowmelt	SBRR, SBGG	1 to 4 events, 2 samples per event	Same package

Phase	Sites	Cadence and timing	What is measured
Extra sites (Full tier only)	SBW_0001, SBW_0002, active fissure (access permitting)	Quarterly	Same package
Isotopes	SBRR, SBGG (subset)	About one sample per site per season with a regular visit	delta-18O, delta-D
QA	All	10% duplicates and blanks	Same package

Table 1: Sample counts by tier.

Tier	Intensive (daily)	Monthly	Storm / melt	Extra sites	QA (10%)	Chemistry samples	Isotope samples
Lean	10	10	0	0	2	22	4
Recommended	14	24	2	0	4	44	8
Full	28	24	8	12	8	80	16

Laboratory cost options

Table 2: Analyte packages and planning prices. These are my estimates, not ALS quotes; replace with the ALS quotation before using the totals in the proposal.

Package	What it includes	Planning price per sample
A: Chloride core	Chloride by IC (EPA 300.0) and alkalinity	$45
B: Major ions (recommended)	A, plus sulfate, fluoride, bromide (IC); Na, K, Ca, Mg, B, Li, Si (ICP); TDS	$135
C: Major ions plus trace elements	B, plus an ICP-MS trace-element suite like the one used on the 2026 field samples	$225
Isotopes	delta-18O and delta-D, outside stable-isotope laboratory	$40

Table 3: Estimated total laboratory cost for one year (chemistry plus isotopes). Excludes shipping, bottles, and field time.

Tier	Chemistry samples	Isotope samples	Package A	Package B	Package C
Lean	22	4	$1,150	$3,130	$5,110
Recommended	44	8	$2,300	$6,260	$10,220
Full	80	16	$4,240	$11,440	$18,640

Recommended option: the Recommended tier with package B, about 44 chemistry samples and 8 isotope samples, roughly $6,260 for the year. It is the smallest design that covers a full seasonal cycle at both stations, includes the ions needed for the mixing-model cross-check, and keeps the daily window short enough to be done in the field by one person. The Lean tier gives a first calibration but no seasonal test; the Full tier adds the fissure and resort wells and a longer daily window, and depends on Ormat access.

Notes for the cost discussion. (1) ALS lab pricing depends on method set, turnaround time and any UNR or project discount, so the totals above scale directly with the package prices in Table 2. (2) Shipping, sample bottles, filters and sampling supplies are not included. (3) Tier totals can be reduced by running packages A or B on routine samples and C only on the quarterly subset.

Equations used for the conductance proxy, model fitting and sampling-error test

1. Specific conductance from logger conductivity. The loggers record raw electrical conductivity EC at water temperature T (degrees C). I remove the temperature effect with the USGS non-linear compensation (coefficient $a = 0.0191$ per degree C, to be recalibrated against field checks):

$$SC_{25} = \frac{EC}{1 + a\,(T - 25)}$$

2. Chloride from specific conductance (the proxy). Chloride is the conservative tracer of thermal water, and in this system conductance tracks chloride because chloride and sodium dominate the thermal fluid. The simplest model is a linear regression, fitted to paired samples:

$$\mathrm{Cl}_t = \beta_0 + \beta_1\, SC_{25,t} + \varepsilon_t$$

The generalized additive model (GAM) adds smooth terms for conductance and temperature and a seasonal harmonic (d is day of year; splines with $k = 4$):

$$\mathrm{Cl}_t = \beta_0 + f_1(SC_{25,t}) + f_2(T_t) + \beta_3 \sin\!\left(\tfrac{2\pi d_t}{365}\right) + \beta_4 \cos\!\left(\tfrac{2\pi d_t}{365}\right) + \varepsilon_t$$

A random forest on the same predictors is the skill benchmark. Models are scored with rolling-origin (blocked, time-aware) cross-validation: each fold trains on all samples up to time t_k and predicts the next block, so autocorrelation cannot inflate skill:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\mathrm{Cl}_i - \widehat{\mathrm{Cl}}_i\right)^2}$$

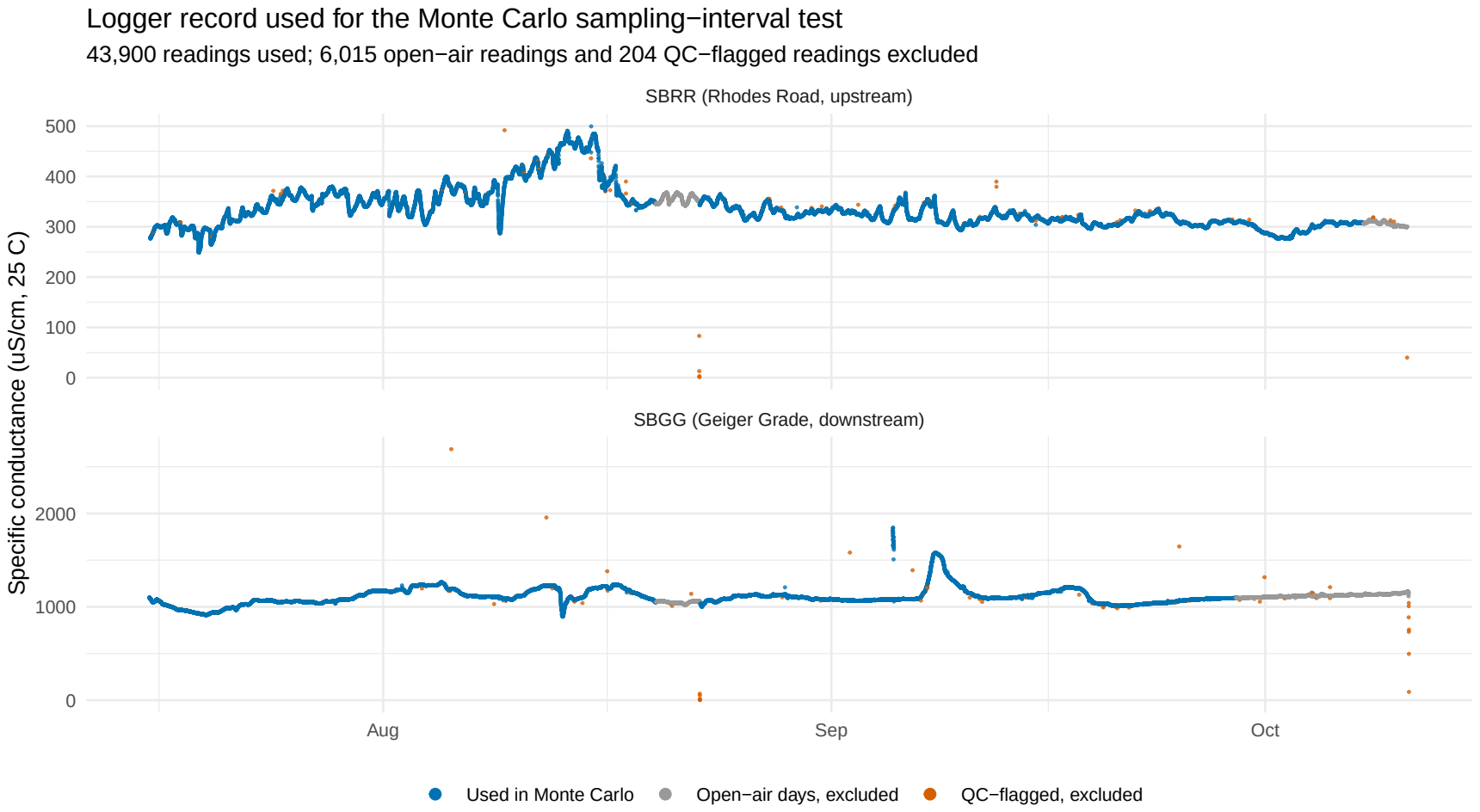
3. Monte Carlo sampling-interval test. For a candidate sampling interval of f days and a logger step of Δt minutes, the number of logger steps between samples is $n_f = f \cdot 1440/\Delta t$. For draw $j = 1, \ldots, 200$ a random start offset $o_j \sim \mathrm{U}\{1, \ldots, n_f\}$ selects samples at steps $o_j, o_j + n_f, o_j + 2n_f, \ldots$. The full series is rebuilt by linear interpolation between those samples, and the error against the dense record y_t is

$$\mathrm{RMSE}_{f,j} = \sqrt{\frac{1}{N}\sum_{t=1}^{N}\left(y_t - \hat{y}_t^{(f,j)}\right)^2}, \qquad \text{reported as } 100\,\frac{\overline{\mathrm{RMSE}_f}}{\bar{y}} \text{ percent of the mean.}$$

4. Chloride flux and thermal discharge. With discharge Q and concentration C, chloride mass flux is the integral of $C\,Q$, computed by the trapezoid rule; the thermal-water inflow between SBRR (upstream) and SBBV (downstream) uses an end-member chloride $C_{TW} = 820$ mg/L (Sorey and Colvard 1992):

$$M = \int C(t)\,Q(t)\,dt, \qquad Q_{TW} = \frac{Q_{BV}\,C_{BV} - Q_{RR}\,C_{RR}}{C_{TW}}$$

Conductivity record and the data each analysis uses



Open−air days: the 3 days before each retrieval (both loggers) and SBGG from 2026−09−29, when it began to be exposed to air (daily temperature range rises from about 5 to 7−8 C). The Monte Carlo test subsamples the blue record at 1, 2, 3, 7 and 14 days and rebuilds it by interpolation (equation 3).

Figure 1

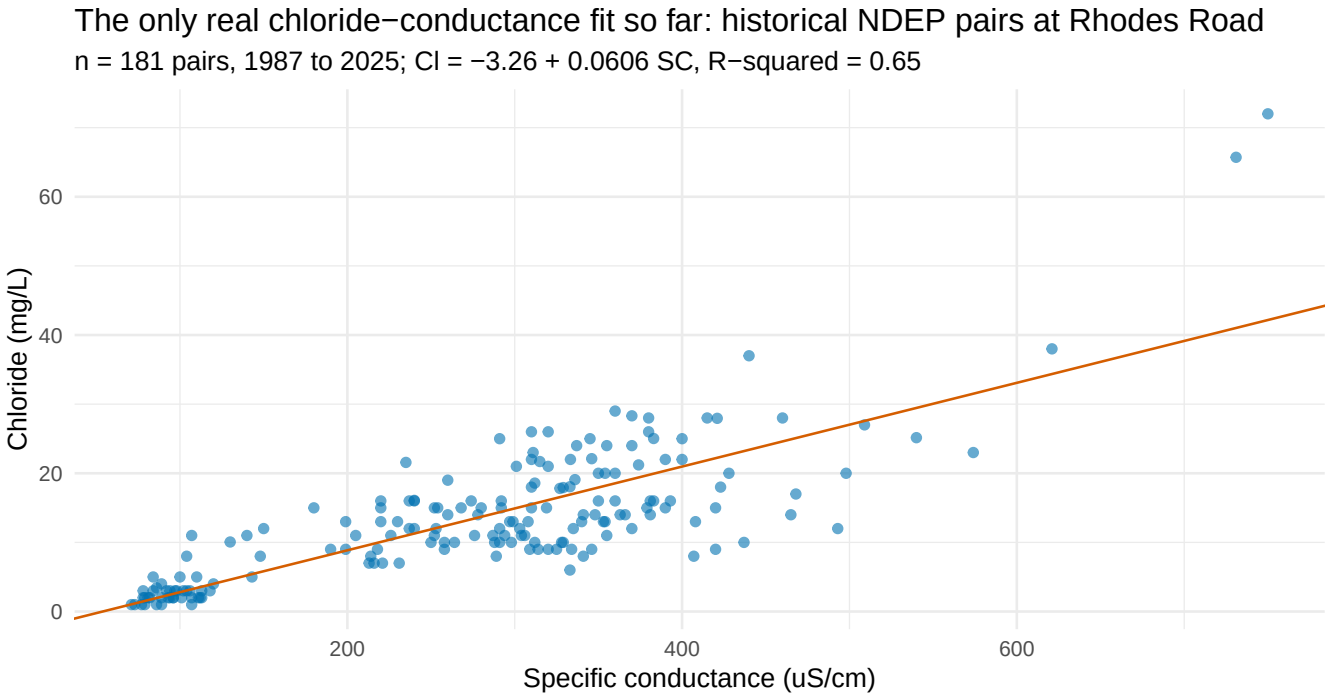

The only real chloride−conductance fit so far: historical NDEP pairs at Rhodes Road
n = 181 pairs, 1987 to 2025; Cl = −3.26 + 0.0606 SC, R−squared = 0.65
Chloride (mg/L)
0
20
40
60
200
400
600
Specific conductance (uS/cm)
No chloride sample yet overlaps the logger record (0 pairs), so the GAM and random forest have only been run on synthetic data.
The calibration campaign proposed above supplies the first real pairs to fit equation 2.

Figure 2

Why the proposed sampling interval matters: error grows with days between samples

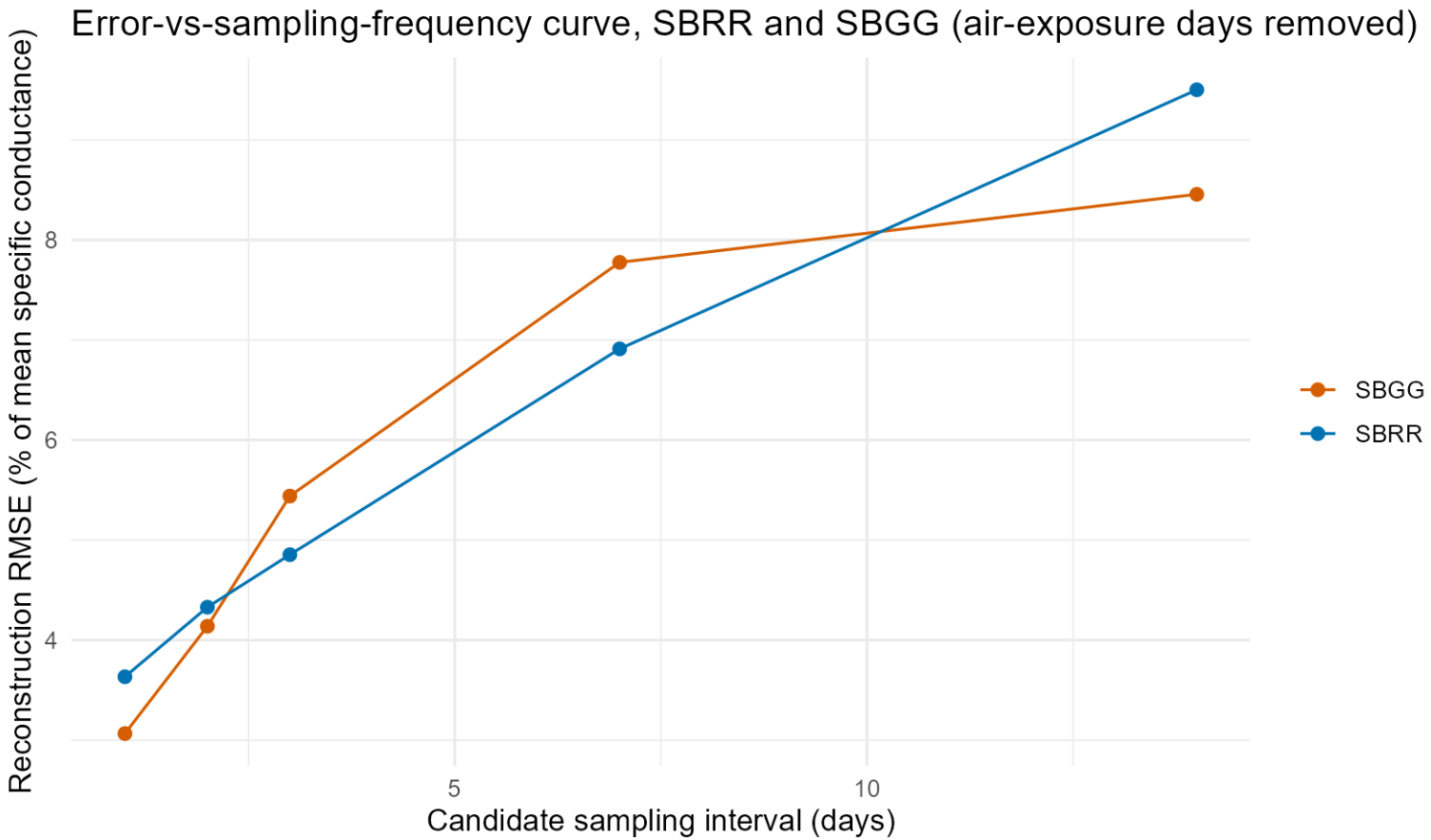


Figure 3: Reconstruction error of the conductance record against the candidate interval between samples, SBRR and SBGG (200 random start days per interval; QC-flagged readings, open-air days before retrieval, and SBGG after 2026-09-29 excluded). Mean error rises from about 3-4% of mean conductance with daily samples to about 8-10% at 14 days.

What the GAM has and has not been run on

The sampling interval in this plan comes from the Monte Carlo test on the **real** logger record (Figures 1 and 3). The chloride models themselves, including the GAM, have **not** been fitted to real paired data, because no chloride sample yet overlaps the logger record (zero pairs). They have been run only as a code test

on a simulated data set (`simulate_synthetic_pairs()` in `03_chloride_models.R`): 60 samples every three days from 1 May 2026, with

$$SC = 300 + 40\sin\!\left(\tfrac{2\pi d}{365}\right) + \mathrm{N}(0, 15), \quad T = 15 + 8\sin\!\left(\tfrac{2\pi(d-60)}{365}\right) + \mathrm{N}(0, 1.5),$$

$$\mathrm{Cl} = 1 + 0.071\,SC_{\mathrm{lag}} + 0.1\,T_{\mathrm{lag}} + \mathrm{N}(0, 1.5),$$

where the lag is six days. The slope 0.071 comes from the single real SBRR pair (Cl 17.1 mg/L at 241 uS/cm on 2026-05-01), so the simulated chloride stays in a realistic range; every other value is made up. Because the data were generated from a known linear rule, the exercise shows only that the fitting and cross-validation code runs and recovers a known relationship. It does not show how well a GAM will perform on Steamboat Creek, and it did not influence the sampling interval. The only real fit is the linear regression on 181 historical NDEP pairs in Figure 2, which predates the loggers. The calibration samples proposed in this document supply the first real pairs for the GAM.